Week 1: Setup, Acquisition & Baseline DEM Generation

- Install and configure the open-source DEM software, LIGGGHTS, to run baseline simulations.
- Download the publicly available training datasets (Sand, SandRamps, and WaterDropSample) from DesignSafe DataDepot (doi: 10.17603/ds2-0phb-dg64).
- Run 50 to 200 LIGGGHTS simulations to generate a cohesive granular compaction dataset under varying stress conditions.
- Vary the applied stress, displacement, and cohesion parameter (using the JKR/DMT model) across the simulation runs.

Week 2: Data Extraction and Formatting

- Extract data regarding particle positions, velocities, and normal and tangential contact forces at each timestep.
- Save the extracted DEM data into the .npz (NumPy compressed) format.
- Ensure each trajectory is stored as a tuple of positions and particle types.

Week 3: Architecture Initialization & Baseline Deployment

- Initialize DeepMind's Graph Network Simulator (GNS) to simulate particle dynamics and predict physical trajectories.
- Validate the baseline architecture using the public DesignSafe dataset.
- Begin building a custom Graph Neural Network (GNN) from scratch using PyTorch Geometric (PyG).

Week 4: Parallel Training

- Train the custom neural network to map initial particle states to final compaction densities and void ratios.
- Simultaneously train the GNS baseline model on the self-generated .npz compaction dataset.

Week 5: Comparative Benchmarking & Initial Iteration

- Benchmark the custom model's performance against the open-source GNS baseline to identify shortcomings.
- Refine the PyG spatial graph convolutions and specialized message-passing operators for learning localized particle-level interactions.

Week 6: Advanced Optimization (Addressing Shortcomings)

- Iterate on the custom model's architecture based on the initial benchmark results to improve its predictive accuracy on final compaction densities.
- Ensure the refined custom model successfully bypasses iterative contact detection algorithms, effectively reducing computation time.

Week 7: Final Evaluation

- Evaluate the optimized custom model's predictive accuracy against the traditional computational outputs.
- Generate comparative plots of force vs. displacement for different materials to benchmark standard DEM results against the ML predictions.

Week 8: Final Compilation & Submission

- Finalize the ML surrogate model bridging computational efficiency and physical accuracy.
- Complete the project reporting and wrap up the work period by the last week of October.